Applied Machine Learning

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ML model training ⟶ data
+
Algorithm
+
computer

☆ ML Solution Framework

Step I - Understanding the business problem

- problem formulation
- understand importance and impact of the problem
- understand return on investment ROI
- Check if AI/ML is really needed for solving this problem or not.

Once we confirm that building ML solution for given problem is valid and profitable then move to the next step.

Domain knowledge of given problem is really helpful in all steps of this framework.

Step 2 - Data collection and management

- confirm properties of data required
  - decide if we need labelled/unlabelled data
  - decide important features, target, dataset size.

- proper data collection
  - maintain high quality of data
  - source of the data must be relevant
  - data shouldn't have missing, null, wrong values

- Data Management
  - manage data security, backups
  - access of data should be seamless

Step 3 - Understand the collected data - EDA

- EDA, univariate, bivariate
- Understand data distributions
  - central tendency, dispersion
  - correlations, trends, comparisons
  - outliers, missing, null values
  - duplicate datapoints

Quality of data is verified in this step.

We confirm if data is good enough for successful ML solution or not.

If data quality is bad, we'll need to perform step 2 again

After step 3 we'll know

- nature of data
- useful features and target
- supervised or unsupervised
- regression or classification or clustering, etc

Step 4 - Data Preprocessing

- Data cleaning
  - handling null values, outliers, invalid values
- data transformation
  - feature scaling
  - feature encoding
- feature engineering

Step 5 - Train Test split of data

ML → data → Algorithm → computer

1000 datapoints

train data 80% = 800

test data 20% = 200

Train data

- used for model training
- seen data

Test data

- used for model testing after training is completed
- unseen data

model evaluation happens on both datasets

Step 6 - ML model training

- training of algorithms
- try out multiple algorithms in this step

Step 7 - Model evaluation, deployment and monitoring

- model evaluation on all parameters
- deployment for end users
- monitoring for errors, performance

We've already covered all concepts to perform steps 1, 2 and 3

Now lets understand step 4

Data Preprocessing

☆ Null / missing values handling

- most of the ML models don't accept null values as input

- To handle null values we can do

Data prunning
- delete data
- remove null value rows / columns

Data imputation
- replace null values with valid values using an assumption